



Tests for carboxylic acids

Carboxylic acids are detectable mainly by their solubility characteristics. They are soluble in both dil. NaOH and NaHCO₃ solutions.

1. Aq. NaHCO₃

Procedure: Dissolve a small amount of compound in cold/hot water and add 5% aqueous NaHCO₃ solution. Observe the solution carefully. If the compound is an acid, bubbles of CO₂ will be seen to form (effervescence).



Phenols usually do not liberate CO₂ from NaHCO₃. However certain phenols are acidic enough to behave much like carboxylic acids. This is especially true of those substituted with electron-withdrawing groups at the ortho and para ring positions. Amine salts, e.g. C₆H₅NH₃⁺Cl⁻, also liberate CO₂ from aqueous NaHCO₃.

2. Esterification

Procedure: Warm a small amount of the compound with a few drops of ethanol and one drop of Conc. H₂SO₄ in a dry test tube. Cool and pour in to a little water contained in a boiling tube. Note the pleasant fruity smell of ester.



3. Ferric chloride test

Neutral solutions should be used for this test. Excess acid prevents the formation of color and/or precipitates. Excess alkali gives a reddish brown precipitate of $\text{Fe}(\text{OH})_3$.

Procedure: Dissolve the solution in water or if insoluble, in a few drops of ethanol and add NH_4OH drop by drop to make the solution just alkaline to litmus. Boil gently until the smell of ammonia is completely removed. Cool this neutralized solution and add neutral ferric chloride. Observe any colorations or precipitates. Warm the mixture and observe any color changes. Cool and add dil. H_2SO_4 .

Preparation of neutral ferric chloride solution

Add NH_4OH drop by drop into solution of FeCl_3 until faint, but permanent precipitate of $\text{Fe}(\text{OH})_3$ is obtained. Filter and use the filtrate, which should be reddish brown in color. Neutral FeCl_3 must be prepared freshly as neutral solution slowly become acidic with time.

Tests for phenols

The pK_a of the typical phenol is 10, whereas pK_a of a carboxylic acid is usually near 5. Hence, phenols are generally not soluble in weak basic NaHCO_3 but do dissolve in NaOH solution, which is more strongly basic. [Note: More acidic phenols, like the nitrophenols dissolve in 10% aq. NaOH solutions forming colored phenoxide anions that have a high degree of conjugation].

Caution! Phenols are caustic action on skin and cause severe burns. Wash with plenty of water with a very dilute solution of bromine water if you spill phenol on your skin.

1. FeCl_3 test

Procedure: Add a few drops of an aq. Solution of neutral FeCl_3 to a dilute aqueous or ethanolic solution of the phenol. Most phenols produce an intense red, blue, purple or green color. Add dil. HCl and observe any color changes.

2. Bromine water test

Procedure: Add a saturated solution of bromine in water to a 1% aq. Solution of the unknown drop by drop with shaking, until the bromine color is no longer discharged. A positive test is indicated by the precipitation of a substitution product (usually turbidity) at the same time that the bromine color is discharged.

3. Phenol in alkaline solution from azo dyes with diazotized aniline.

Procedure for this test is given under the tests for amines.

Tests for Amines

Amines are detected best by their solubility behavior and their basicity. If the unknown is soluble in water, the aqueous solution will have a high pH. Amines dissolve in dil. HCl forming

the amine hydrochloride. Treatment with NaOH solution liberates the free amine, which separates as oily drops or as a solid from the alkaline solution.



1. Carbylamines test (Isocyanides formation)

Primary amines react with chloroform and alcoholic KOH to form volatile isocyanides which are recognized by their unpleasant odors.



Amphoteric amino compounds (eg. Amino carboxylic acids and amino phenols) give potassium salts of the isocyanides, which may not give a smell as they are non-volatile. N-Substituted amides like acetanilide are hydrolyzed by the alkali to primary amines, which will answer the test.

Procedure: add a few drops of CHCl_3 and alcoholic KOH solution (2ml) to a small amount of the compound. Mix well and gently warm for a few minutes. Odor of carbylamines is detected easily when a few drops are added to a boiling tube containing water.

Caution! Isocyanides are extremely poisonous. Immediately after this test is completed, cool and carefully add a few drops of conc. HCl to hydrolyze the isocyanide to amine, which are odorless.

2. Nitrous acid test

Primary aliphatic amines react with HNO_2 liberating N_2 and primary aromatic amines form diazonium salts which are stable at $0-5^\circ\text{C}$ and can be coupled with phenols to give azo dyes. Secondary amines form nitroso derivatives.

Procedure: Dissolve the unknown (0.1 g) in water (2 ml) to which conc. H_2SO_4 (6-8 drops) have been added. Cool the solution to $0-5^\circ\text{C}$ in an ice bath. Also cool 2 ml of aq. NaNO_2 in another test tube. In a 3rd test tube, prepare a solution of 3-naphthol (0.1 g) in aq. NaOH (2 ml) and place it in an ice bath to cool.

Add the cold NaNO_2 solution, drop by drop with shaking, to the cooled solution of the amine. Look for bubbles of N_2 gas.

Substantial evolution of gas at $0-5^\circ\text{C}$ indicates a primary aliphatic amine, RNH_2 . The formation of yellow oil or a yellow solid usually indicates a secondary amine, R_2NH . Tertiary amines do not react. If little or no gas evolves at 5°C , take half the solution and warm it gently to room temperature. Nitrogen gas bubbles at this elevated temperature indicate that the original compound was a primary aromatic amine, ArNH_2 . Add the remaining diazo solution to the solution of β -naphthol in alkali. If a red dye precipitates, the unknown has been conclusively shown to be a primary aromatic amine.

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